Royal Military College of Canada

Fall / Winter - Exam Cover Sheet

(Please circle appropriate term above)

Course Code(s): CSE101 **Section #(s):** 01,02,03

Date of the Exam: 17-Dec-19 **AM** ☑ **PM** ☐

Instructor(s) Name(s): Diane Kelly, Rene Sturgeon, Jason Zwicker, Francois Rivest

Length of Exam: 3.0 hrs

Student Name:

Student #:_____

INSTRUCTIONS:

Exam	**Exam Type:** Closed Book
	Answer booklet: No Booklet / Answers Directly on the Exam **Multiple Choice Answer Sheet Required** ☐
Reference sheets	**Details :** Single Sided Handwritten Sheet (8.5x11) **Other Specifications:** Click here to enter text.
Reference Material(s) permitted:	**Book/Manual:** Choose an item. **Book/Manual:** Choose an item. **Book/Manual, list here:** Click here to enter text. **Book/Manual, list here:** Click here to enter text.
Calculators:	Not Permitted

Exam Additional Instructions:

i.e. Explain in detail your answers and use back side of exam pages for rough draft.

For Undergraduate Registrar (UG) Office Use Only (Instructor's Response Required):

Will this exam copy be combined with another course code or section? Yes ☒ **No** ☐ **If yes, which course code or section(s): CSE101 All 3 sections**
Exam is translated in English & French (on Same Copy) : Yes ☐ **No** ☒
May this exam be published once exam period is over? : Yes ☒ **No** ☐

Note: Print, date & sign, then photocopy and attach to each exam as the cover. The UG office MK453 will require your official copy of this cover sheet by mid-month in month prior to start of exams.

Instructor's Signature confirming above information is accurate: _____

Date:_____

ROYAL MILITARY COLLEGE OF CANADA
FALL TERM EXAMINATIONS 2019
FIRST YEAR EXAMINATION

CSE101: Introduction to Algorithms and Computing

December 17 2019 **Time: 3 Hours**

Examiner: Dr. Diane Kelly, Dr. Francois Rivest, Dr. Rene Sturgeon, Maj. Jason Zwicker

Notes:

1. Print your name and college number in the spaces below.
2. Circle the name of your instructor.
3. Answer the questions in the spaces provided. Do not detach pages from the examination.
4. Use the back (left side) of the examination pages for rough work. No rough work on the back (left side) examination pages will be marked unless specifically indicated by the student.
5. This is a closed book examination. A one page hand-written single-sided aide-memoire is allowed.
6. A list of several MATLAB functions complete with a short description of each is provided on the last page of the exam.
7. Any electronic devices capable of calculations or programming are forbidden.
8. In case of doubt as to the meaning of a question, clearly state your assumption(s) and proceed.

NAME: _____ College Number: _____

Turn over for question 1.

EXAMINER'S USE ONLY

Question Number	Value	Mark
1	6	
2	4	
3	5	
4	2	
5	4	
6	4	
7	5	
8	5	
9	5	
10	2	
11	4	
12	5	
13	4	
14	5	
Total	60	

Question 1. [6 points] For each of the following, write your answer in the space provided.

a) A student in CSE101 typed the following:

```
S = '25 Main Street Kingston';
```

Realizing the address number was wrong, the student typed

```
S(1) = 33;
disp(S)
```

The output from the disp instruction is

```
!5 Main Street Kingston
```

Explain what the student did wrong and why the ! character was displayed.

b) A programmer calculates a timer by repeatedly adding a tenth of a second:

```
time = 0;
for k = 1:100000
  time = time + 0.1
end
```

At some point later in the program, the programmer does the following check:

```
if time == 10000
  disp('time to launch missile')
end
```

The programmer never sees this message because of a type of error seen only with floating point number calculations. Explain why this floating point error occurs.

c) Describe two of the activities carried out by the Central Processing Unit (CPU) of a Von Neumann computer architecture.

Question 2. [4 points]

The following is MATLAB code for a sort algorithm. In the space below, show what is output by the disp instruction when this code is executed on the vector (array) V:

```
V = [42  71  26  16  82  53];
```

MATLAB code:

```
N = length(V);
sorted = 0;
while ~sorted
  sorted = 1;
  for k = 1:N-1
    if  V(k) < V (k+1)
      hold = V(k);
      V(k) = V(k+1);
      V(k+1) = hold;
      sorted = 0;
    end
  end
  disp(V)
end
```

Show what is displayed here:

Question 3. [5 points]

The following function is intended to find the largest value in the main diagonal of a square matrix. You are given the function statement and a description of the input and output. Complete the function.

```
function [pos, val] = FindBigDiag(M)
%
% Input: M is a square matrix, that is
%            the number of rows equals the number of columns
%
% Output: pos is the position of the largest value in
%                 the main diagonal of M
%           val is the value found at pos
%
```

Question 4. [2 points]

Write MATLAB code to do the following:

Ask your user for a value and store it in the variable x.

Take the square root of the absolute value of x and store the answer in the variable output.

Display this answer.

Question 5. [4 points] For each of the following, give the answer in the space provided.

a) Convert 123 base 10 into its binary representation stored in an unsigned 8-bit word.

b) Convert 00101101 base 2 into its base 10 representation.

c) Convert –17 base 10 into its 8-bit base 2 representation where it is stored as signed 2s complement.

d) The binary number 11101101 is stored as 8-bit signed 2s complement. What is its representation in base 10?

Question 6. [4 points]

What is the output generated by the following MATLAB code:

```
count = 0;
total = 0;
a = 0;
for i = 1:3:20
    total = total + i;
    count = count + 1;
    if rem(i,2) & i>a
        a = i;
    end
end
fprintf('The average is %.2f.\n', total / count);
fprintf('The other value is %d.\n',a);
```

Write your answer here:

Question 7. [5 points]

a) Write MATLAB code to do the following:

Create a vector (array) with the values 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Use a `for` loop to check each entry of the vector: if the value of the entry is even, set that entry to zero. (Note: `r = rem(N, d)` gives the remainder for N/d)

Display the vector after the `for` loop is completed.

b) Show what is output by the final display statement of your code.

Question 8. [5 points]

Write a function triplerandi that generates three different random numbers that are within a given range [low, high]. Note: You can assume that the [low, hi] range spans at least three integers.
For example, [n1,n2,n3] = triplerandi(1,10) could generate values of n1=5,n2=10,n3=4 because they are all different and within the range [1,10]. It will not output values such as n1=4,n2=6,n3=4; because n1 and n3 are equal.

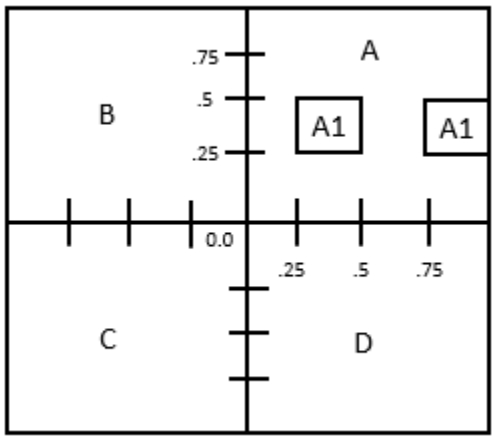
Note: the MATLAB function randi can be used as N = randi(L,H)
where N is an integer in the range $L \leq N \leq H$

Complete the function triplerandi starting with this MATLAB code:

```
function [val1,val2,val3]=triplerandi(low,hi)
%
% This function randomly generates three different integers in
% the given range of low and high.
% Inputs:
%   low: the lowest integer that can be generated
%   hi: the highest integer that can be generated
% Outputs:
%   val1,val2,val3 : three different integer values
%                    within given range
```

Question 9. [5 points]

A student has written a function **classify** that takes a point (x,y) and displays the name of the region where that point falls according to the following diagram:



For example, if the point is x = -0.5, and y = -0.5 then this function displays 'C' because the point falls in the region C.

For example, if the point is x = 0.3, and y = 0.3 then this function displays 'A1' because the point falls in one of the regions A1. 'A' does not need to be displayed in that case.

Finally, consider a value of zero to be positive (or non-negative).

On the **following page** is the code the student has written, but it does not work. Help the student by fixing his or her code to make it correct. Each disp statement is correct, but you need to verify that each if condition is correct. You must fix all the errors (write the corrected line below the incorrect condition.).

Question 9. [continued]

```
function [] = classify(x,y)
%
% This function displays the classification of the point (x,y)
% according to regions on a known map.  (See diagram.)
%
% Input: x,y coordinates of a point in the range [-1,1]
% Output: None. Display the region where the point falls:
%             A, A1, B, C, D

if (x >= 0) && (y >= 0)        % Test for A


        if (x >= .25 && x <= .50) && (y >= 0.25 && y >= 0.50)


                disp('A1');

        elseif (x >= .75 && x <= .75) && (y >= 0.25 || y <= 0.50)


                disp('A1');

        else


                disp('A');

        end

elseif (x < 0) || (y >= 0)      % Test for B


        disp('B');

elseif (x < 0) && (x < 0)      % Test for C


        disp('C');

else                            %  Test for D


        disp('D');

end
```

Question 10. [2 points]

a) Given the following MATLAB code, what is the value of x? Choose the correct answer.

```
x = '6' + 4;
```

i) x = '10 '

ii) x = 10

iii) x = 58

b) Given that the ASCII decimal number for character 'm' is 109, and for character 'n' is 110, what is the ASCII decimal number for character 's'?

Question 11. [4 points]

For each of the following, determine the value of the Boolean expression. Show your work.

a) If x has the value 0, what is the value of the following?

(x > 0) & (x <= 10)

b) If x = 3, y = 11 and z = 1 ,what is the value of the following?

~((x^3 > 2*y) & z)

Question 12. [5 points]

Write MATLAB code for a function, sumLargePerColumn , that computes a special sum for each column of a matrix (2 dimensional array). The sum only includes those values in the column that are larger than or equal to 100 or smaller than or equal to -100. Store all the sums in a vector (array), newArray, where the sum of column 1 is in position 1 of newArray, the sum for column 2 is in the second position of newArray, etc. Use for loops to solve this problem. Use comments to explain your code.

As an example, for the matrix:

101	-9	718
-3	12	0
23	-500	1
109	55	-20
-22	33	-140

The function sumLargePerColumn returns:

210	-500	578

Following is the first line of your function. Complete this function.

```
function newArray = sumLargePerColumn (anArray)
```

Question 13. [4 points] In the space provided, write a piece of MATLAB code to do the following:

a) Given two variables First_name and Last_name, create a variable called Name to hold the contents of the first two variables with a space between.

For example:
First_name = 'Angelina';
Last_name = 'Jolie';

After your code is executed, Name has the content 'Angelina Jolie'

Write your code for any names in First_name and Last_name:

b) Undo the code in part (a), that is, given a full name in variable Name, create the variables First_name and Last_name with the respective parts of the contents of Name.

For example, Name = 'Angelina Jolie'

After your code is executed,

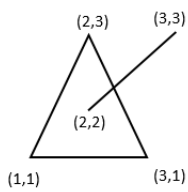
First_name has the content 'Angelina'
Last_name has the content 'Jolie'

Question 14. [5 points]

You are **given** the following function that calculates whether or not two line segments intersect. The function description is as follows:

```
function [answer] = check(pt1x,pt1y,pt2x,pt2y,pt3x,pt3y,pt4x,pt4y)
%
% This function returns true if the line segments lineseg1 and
%    lineseg2 intersect where:
%       lineseg1 has end points pt1x,pt1y and pt2x,pt2y;
%       lineseg2 has endpoints pt3x,pt3y and pt4x,pt4y
% Inputs:
%       pt1x,pt1y and pt2x,pt2y are the endpoints of the first
%                          line segment
%       pt3x,pt3y and pt4x,pt4y are the endpoints of the second
%                          line segment
% Outputs: answer : True (1) if lineseg1 intersects lineseg2;
%                           False (0) if there is no intersection
%
```

Your task: **<u>use</u>** the above function to write your own function, polyintersect, that checks to see if there is an intersection between any given line segment and any of the line segments of a given convex polygon.



For example: If we have the following problem:

$$\text{poly} = \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 2 & 3 \end{bmatrix} \text{ and lseg} = [2 \quad 2 \quad 3 \quad 3]$$

where each row of poly contains the (x,y) coordinates of the vertices of a polygon in order and lseg has the (x,y) coordinates of the endpoints of a line segment, your function returns true because the line segment in lseg intersects one of the line segments of the polygon in poly.

The MATLAB code on the following page gives the start of your new function, polyintersect. Complete this function.

Question 14. [continued]

```
function [intersect] = polyintersect(poly, lseg)
%
% This function calculates whether or not a given line segment
% intersects any of the line segments of a given polygon.
%
% Inputs:
%       poly : an array (matrix) of points representing the
%          vertices of a convex polygon. The first column holds
%          the x values of each vertex, and the second column
%          holds the y values.
%       lseg: a vector representing both end points of a line
%         segment, ie [x1,y1,x2,y2]
% Output:
%       intersect: set to true (1) if the line segment intersects
%             any of the line segments of the polygon;
%             set to false (0) otherwise.
%
```

End of exam

CSE/INF 101 Auxiliary Information Sheet
CSE/INF 101 Fall Final Exam – 17 December 2019

Extra MATLAB functions that may be used on this exam

Explanations

y = abs(x)	y equals the absolute value of x
y = ceil(x)	y is the least integer greater than or equal to x
y = fix(x)	y is the same as floor for positive x and the same as ceil for negative x
y = floor(x)	y is the greatest integer less than or equal to x
y = rand	y is a randomly generated number between zero and one, i.e., in the range (0,1)
y = rem(x, r)	y is the remainder of the integer division x/r
y = sqrt(x)	y is the square root of x
y = strcmp(str1, str2)	y equals 1 if the text strings srt1 and str2 are identical; y equals 0 otherwise
y = strfind(str, s1)	y is a vector with the locations of the first letter of every text string s1 found in text string str; y is an empty vector if s1 not found in str
y = strrep(str, s1, s2)	y is a text string equal to str with every occurrence of text string s1 replaced with text string s2